

TECHNICAL PROPOSAL AND SOFTWARE REQUIREMENTS AND DESIGN DOCUMENT

# Hadia Gifting Registry

Gift registry, item reservation and M-Pesa cash gifting platform with automated escrow and payouts.

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PREPARED FOR

Hadia  
Attn: Tracey, Founder

PREPARED BY

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DATE

6 October 2026

DELIVERY PERIOD

12 weeks

INVESTMENT

KES 610,000

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# Document control

| Version | Date           | Author                    | Description              |
|---------|----------------|---------------------------|--------------------------|
| 1.0     | 6 October 2026 | Codevertex Africa Limited | Issued for client review |

| Role           | Party      | Responsibility                                                   |
|----------------|------------|------------------------------------------------------------------|
| Product owner  | Hadia      | Scope approval, business decisions, legal sign-off, pilot events |
| Delivery lead  | Codevertex | Planning, scope control, reporting, risk management              |
| Technical lead | Codevertex | Architecture, code quality, security, releases                   |

## Contents

|          |                                     |           |
|----------|-------------------------------------|-----------|
|          | <b>Executive summary</b>            | <b>3</b>  |
| <b>1</b> | <b>Introduction</b>                 | <b>4</b>  |
| 1.1      | Purpose                             | 4         |
| 1.2      | Product scope                       | 4         |
| 1.3      | Definitions                         | 4         |
| 1.4      | References                          | 5         |
| <b>2</b> | <b>Overall description</b>          | <b>6</b>  |
| 2.1      | Product perspective                 | 6         |
| 2.2      | Product functions                   | 6         |
| 2.3      | User classes                        | 6         |
| 2.4      | Operating environment               | 6         |
| 2.5      | Constraints                         | 6         |
| 2.6      | Assumptions and dependencies        | 7         |
| <b>3</b> | <b>Functional requirements</b>      | <b>8</b>  |
| <b>4</b> | <b>Non-functional requirements</b>  | <b>10</b> |
| <b>5</b> | <b>Solution architecture</b>        | <b>11</b> |
| 5.1      | Components                          | 11        |
| 5.2      | Architecture principles             | 12        |
| <b>6</b> | <b>Funds management and payouts</b> | <b>13</b> |
| 6.1      | Custody                             | 13        |
| 6.2      | Registry escrow account             | 13        |
| 6.3      | Payout configuration                | 14        |
| 6.4      | Payout process                      | 15        |
| 6.5      | Commission                          | 15        |
| 6.6      | Reconciliation                      | 16        |
| <b>7</b> | <b>Process workflows</b>            | <b>17</b> |
| 7.1      | Hadia business onboarding           | 17        |

|           |                                                   |           |
|-----------|---------------------------------------------------|-----------|
| 7.2       | Registry owner onboarding .....                   | 18        |
| 7.3       | Item listing .....                                | 18        |
| 7.4       | Sharing .....                                     | 19        |
| 7.5       | Item reservation .....                            | 19        |
| 7.6       | Cash contribution .....                           | 20        |
| 7.7       | Registry management .....                         | 20        |
| 7.8       | Automated payout .....                            | 20        |
| 7.9       | Commission and sweep .....                        | 21        |
| 7.10      | Premium upgrade .....                             | 21        |
| 7.11      | Cancellation and refunds .....                    | 21        |
| 7.12      | Exceptions and reconciliation .....               | 21        |
| 7.13      | Thank-you tracking .....                          | 21        |
| 7.14      | Data subject requests .....                       | 22        |
| <b>8</b>  | <b>Data design .....</b>                          | <b>23</b> |
| 8.1       | Scalability measures .....                        | 23        |
| <b>9</b>  | <b>Interfaces, events and notifications .....</b> | <b>24</b> |
| 9.1       | hadia-api endpoints .....                         | 24        |
| 9.2       | Treasury endpoints .....                          | 24        |
| 9.3       | Events .....                                      | 24        |
| 9.4       | Notifications .....                               | 25        |
| <b>10</b> | <b>Reporting and analytics .....</b>              | <b>26</b> |
| <b>11</b> | <b>Security .....</b>                             | <b>27</b> |
| <b>12</b> | <b>Legal and regulatory compliance .....</b>      | <b>28</b> |
| 12.1      | Roles and responsibilities .....                  | 28        |
| 12.2      | Applicable law .....                              | 28        |
| 12.3      | Compliance checklist .....                        | 29        |
| 12.4      | Website content review .....                      | 30        |
| 12.5      | Record retention .....                            | 31        |
| 12.6      | Legal deliverables .....                          | 31        |
| <b>13</b> | <b>Project delivery .....</b>                     | <b>32</b> |
| 13.1      | Methodology .....                                 | 32        |
| 13.2      | Project team .....                                | 32        |
| 13.3      | Implementation schedule .....                     | 33        |
| <b>14</b> | <b>Testing and acceptance .....</b>               | <b>34</b> |
| <b>15</b> | <b>Commercial terms .....</b>                     | <b>35</b> |
| 15.1      | Fixed price .....                                 | 35        |
| 15.2      | Payment schedule .....                            | 35        |
| 15.3      | Monthly running costs .....                       | 35        |
| <b>16</b> | <b>Risks and mitigations .....</b>                | <b>36</b> |
| <b>17</b> | <b>Post-launch support .....</b>                  | <b>37</b> |
| <b>18</b> | <b>Acceptance .....</b>                           | <b>38</b> |

# Executive summary

Codevertex Africa Limited proposes to design, build and launch the Hadia Gifting Registry, a mobile-first platform on which registry owners list gifts for an occasion, share a single link, and receive item reservations and M-Pesa cash contributions from guests. Cash contributions are held in escrow and paid out to owners automatically, net of Hadia's commission, under rules configured by Hadia.

The platform is built on Codevertex's production services for identity, payments, accounting, escrow, approvals and messaging. The registry service, web application and administration console are developed specifically for Hadia.

## Key outcomes

- Registries published and shared within five minutes of sign-up, with no application download for owners or guests.
- Duplicate gifts eliminated through atomic item reservations visible to all guests in real time.
- M-Pesa contributions credited exactly once and reconciled hourly against the PayHero wallet.
- Payouts and commission deductions executed automatically by threshold and event date, with approval required only above Hadia's limit.
- Commission transferred to Hadia's business account daily without manual intervention.
- Compliance with the Data Protection Act, consumer protection, payments and accessibility requirements from launch.

## Engagement summary

|                                    |                                   |                                                     |                                         |
|------------------------------------|-----------------------------------|-----------------------------------------------------|-----------------------------------------|
| <b>12 weeks</b><br>Delivery period | <b>KES 610,000</b><br>Fixed price | <b>KES 13,000 to 26,000</b><br>Monthly running cost | <b>4 milestones</b><br>Payment schedule |
|------------------------------------|-----------------------------------|-----------------------------------------------------|-----------------------------------------|

## SECTION 1

# Introduction

## 1.1 Purpose

This document specifies the requirements, design, delivery plan and commercial terms for the Hadia Gifting Registry. It is the basis for development, testing and acceptance.

## 1.2 Product scope

| Area              | Included                                                                                                                         |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Public website    | Landing page, how it works, occasions, help centre, legal pages, cookie consent                                                  |
| Owner application | Mobile sign-up, registries, items, sharing, gift log, thank-you tracking, payout settings, premium upgrade, privacy settings     |
| Guest experience  | Registry page, reservations, M-Pesa contributions (Airtel Money and card where enabled), receipts, reservation cancellation      |
| Funds management  | Escrow per registry, automated payouts, commission deduction and sweep, contribution limits, refunds, reconciliation, accounting |
| Administration    | Registry and user management, approvals, exception queue, reports, fee and limit settings, moderation, data requests             |
| Messaging         | SMS and WhatsApp receipts and alerts, email, marketing opt-in                                                                    |
| Compliance        | Legal policy drafts, consent records, data export and deletion, WCAG 2.2 AA accessibility                                        |

**Excluded:** native iOS and Android applications, wedding planning and RSVP tools, gift delivery and retail, retailer affiliate integrations, formal legal opinions, marketing campaigns.

## 1.3 Definitions

| Term           | Definition                                                                    |
|----------------|-------------------------------------------------------------------------------|
| Registry       | An owner's list of items and cash funds for an event                          |
| Escrow account | The treasury record holding a registry's cash contributions                   |
| Payout         | Transfer of a registry balance, net of commission, to the owner               |
| Commission     | Hadia's percentage fee on cash contributions, deducted at payout              |
| Sweep          | Scheduled transfer of earned commission to Hadia's business account           |
| Premium        | One-time paid upgrade per registry                                            |
| STK push       | M-Pesa payment prompt sent to the payer's phone                               |
| PayHero Team   | Hadia's PayHero account with an isolated wallet                               |
| KYC, AML       | Know Your Customer; anti-money laundering                                     |
| ODPC, DPIA     | Office of the Data Protection Commissioner; data protection impact assessment |
| eTIMS          | KRA electronic Tax Invoice Management System                                  |

|          |                                                   |
|----------|---------------------------------------------------|
| PWA      | Progressive web application                       |
| UAT      | User acceptance testing                           |
| RPO, RTO | Recovery point objective; recovery time objective |

## 1.4 References

- Hadia landing page concept and Gift Wishlist and Cash-Gifting Platform build report, August 2026
- Codevertex platform documentation: treasury escrow, PayHero integration, settlement flows, authentication, notifications
- Data Protection Act, 2019 and Data Protection (General) Regulations, 2021
- Web Content Accessibility Guidelines 2.2; OWASP Application Security Verification Standard 4

## SECTION 2

# Overall description

## 2.1 Product perspective

Hadia operates as a tenant of the Codevertex platform. Two new applications, hadia-ui and hadia-api, provide the registry product. Identity, payments, ledger, escrow, approvals and messaging are provided by existing platform services, described in [Solution architecture](#).

## 2.2 Product functions

- Registry creation and management for weddings, baby showers, housewarmings, graduations and birthdays
- Item listing by manual entry or product link, with reserve, fund or combined modes
- Guest reservations without duplicates, and cash contributions by M-Pesa
- Escrow of contributions with automated payouts and commission deduction
- Premium upgrade per registry: custom link, printable list, instant payouts
- Administration, reporting, reconciliation and compliance management

## 2.3 User classes

| Role                | Description                                       | Permissions                                                                                       | Authentication                 |
|---------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------|--------------------------------|
| Guest               | Family, friends, diaspora relatives, colleagues   | View registry, reserve, contribute, leave a message, cancel own reservation                       | None; OTP when reserving       |
| Registry owner      | Couple, parent, graduate or host, aged 18 or over | Manage registries and items, share, view gifts, thank guests, set payout number, purchase premium | Mobile number and OTP          |
| Co-owner            | Invited by the owner                              | Edit items, thank guests                                                                          | Mobile number and OTP          |
| Hadia support       | Hadia staff                                       | User support, moderation, data requests                                                           | SSO with MFA                   |
| Hadia finance       | Hadia staff                                       | Approvals, ledger, reports, fees and limits                                                       | SSO with MFA; OTP per approval |
| Hadia administrator | Founder and delegates                             | All permissions, staff roles, settings                                                            | SSO with MFA                   |

## 2.4 Operating environment

- Guests and owners: any modern mobile or desktop browser; installable as a PWA
- Hosting: Codevertex Kubernetes platform with PostgreSQL, Redis, NATS JetStream and object storage, behind Cloudflare
- Payments: PayHero (M-Pesa, Airtel Money, card); messaging: Africa's Talking, WhatsApp, Brevo

## 2.5 Constraints

- Funds are collected, held and disbursed only through a licensed payment service provider.
- The registry service does not move money; all payments are executed by the treasury service.
- Retailer product data is inconsistent; manual item entry is the primary method.

- Payment onboarding depends on Hadia's company registration and PayHero KYC.

## 2.6 Assumptions and dependencies

- Hadia is a registered Kenyan company with a KRA PIN and business bank account or paybill by week 2.
- Hadia completes PayHero KYC to tier 3 and provides requested documents within 3 working days.
- Hadia appoints a product owner with authority to decide within 2 working days.
- Hadia provides brand assets, approves copy and supplies consented testimonials.
- Hadia engages a Kenyan advocate to approve the legal pack and advise on payment regulation.
- Hadia recruits 3 to 5 pilot events for week 11 and registers the domain name.
- Hadia subscribes to the Codevertex plan with escrow and M-Pesa features, included in running costs.

## SECTION 3

# Functional requirements

Priority: **M** must have, **S** should have, **C** could have.

| ID                           | Requirement                                                                                                                                                                    | Pri. |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| ACCOUNTS AND ONBOARDING      |                                                                                                                                                                                |      |
| FR-01                        | Owners sign up and sign in with a mobile number and a 6-digit one-time password sent by SMS, with WhatsApp as fallback. Codes expire after 5 minutes. Sending is rate limited. | M    |
| FR-02                        | Owners confirm they are 18 or over and accept the terms and privacy notice. Version and timestamp are recorded.                                                                | M    |
| FR-03                        | Owners verify the payout M-Pesa number by one-time password. The registered M-Pesa name on the first payout is matched against the owner's name; mismatches are flagged.       | M    |
| FR-04                        | Owners invite co-owners by mobile number. Co-owners cannot change payout details.                                                                                              | S    |
| REGISTRIES AND ITEMS         |                                                                                                                                                                                |      |
| FR-05                        | Create a registry with type, title, date, optional location, cover photo and description. An owner may hold multiple registries.                                               | M    |
| FR-06                        | Add items manually: title, description, price, quantity, photo, store name and mode (reserve, fund or both).                                                                   | M    |
| FR-07                        | Add items from a product link. Public page metadata prefills the form for owner confirmation.                                                                                  | S    |
| FR-08                        | Create cash funds with an optional target.                                                                                                                                     | M    |
| FR-09                        | Reorder, edit, hide and archive items. Items with gifts can be archived but not deleted.                                                                                       | M    |
| FR-10                        | Registry visibility: public, unlisted (default) or passcode protected.                                                                                                         | S    |
| SHARING AND GUEST EXPERIENCE |                                                                                                                                                                                |      |
| FR-11                        | Each registry has a short link with a preview image, title and description for WhatsApp, Facebook and SMS.                                                                     | M    |
| FR-12                        | Guests view items, reservation status, fund progress and the event description without an account.                                                                             | M    |
| FR-13                        | Reservations are atomic; the last unit of an item cannot be reserved twice.                                                                                                    | M    |
| FR-14                        | Guests cancel their reservation from a link sent to them.                                                                                                                      | M    |
| FR-15                        | Owners control whether reserving guests' names are visible to other guests. Hidden by default.                                                                                 | S    |
| CONTRIBUTIONS AND PAYOUTS    |                                                                                                                                                                                |      |
| FR-16                        | Guests contribute toward an item or fund by M-Pesa STK push, within configured limits. Airtel Money and card are available where enabled.                                      | M    |
| FR-17                        | Guests are charged the amount entered. No fee is added to guest payments.                                                                                                      | M    |
| FR-18                        | Where the STK push cannot be delivered, guests are offered a paybill and account number.                                                                                       | S    |
| FR-19                        | Guests receive an SMS or WhatsApp receipt. Owners receive a gift notification.                                                                                                 | M    |
| FR-20                        | Payouts run automatically when the balance reaches the threshold and after the event date. Premium registries pay out on each contribution.                                    | M    |

|                       |                                                                                                                                    |          |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>FR-21</b>          | Commission is deducted automatically at payout and itemised on every payout statement.                                             | <b>M</b> |
| <b>FR-22</b>          | Earned commission is transferred automatically to Hadia's business account on a schedule, above a minimum amount.                  | <b>M</b> |
| <b>FR-23</b>          | Payouts at or above the approval limit require finance approval with OTP. Payouts below the limit run without intervention.        | <b>M</b> |
| <b>FR-24</b>          | Cancelling a registry refunds each contribution to the paying number. Contributions without a number are listed for manual refund. | <b>M</b> |
| <b>FR-25</b>          | Owners download a statement of contributions and payouts in PDF or spreadsheet format.                                             | <b>S</b> |
| <b>OWNER TOOLS</b>    |                                                                                                                                    |          |
| <b>FR-26</b>          | Gift log with guest name, amount, item and message; thank-you status; WhatsApp thank-you message.                                  | <b>M</b> |
| <b>FR-27</b>          | Premium upgrade per registry: custom link, printable list, instant payouts.                                                        | <b>M</b> |
| <b>FR-28</b>          | Reminders to guest and owner 14 days after an unpurchased reservation. Owner may release the item.                                 | <b>C</b> |
| <b>ADMINISTRATION</b> |                                                                                                                                    |          |
| <b>FR-29</b>          | Search registries, owners and contributions. View a registry as a guest.                                                           | <b>M</b> |
| <b>FR-30</b>          | Configure commission, premium price, contribution limits, payout thresholds, approval limit and sweep schedule.                    | <b>M</b> |
| <b>FR-31</b>          | Exception queue: failed payouts, name mismatches, manual refunds, reconciliation variances, reported registries.                   | <b>M</b> |
| <b>FR-32</b>          | Content moderation with recorded reasons and owner notification.                                                                   | <b>M</b> |
| <b>FR-33</b>          | Reports: contributions and value by period, active registries, commission, premium sales, conversion, payout times.                | <b>M</b> |
| <b>FR-34</b>          | Data subject requests (access, correction, deletion) with deadline tracking.                                                       | <b>M</b> |

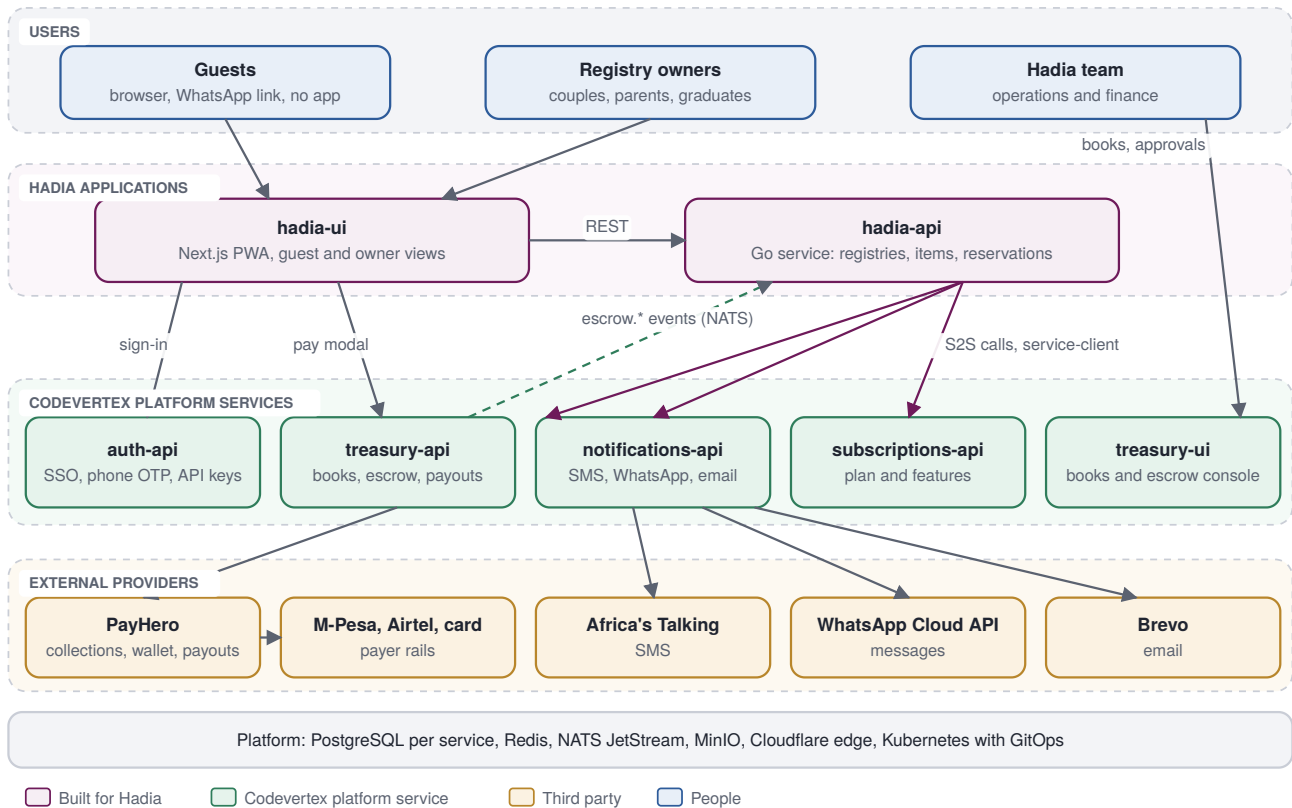
## SECTION 4

# Non-functional requirements

| ID            | Attribute                  | Requirement                                                                                                                                                                 |
|---------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NFR-01</b> | <b>Mobile performance</b>  | Guest page initial transfer under 150 KB excluding photos. Largest Contentful Paint under 2.5 seconds on a mid-range Android device over 3G. Photos served as resized WebP. |
| <b>NFR-02</b> | <b>Response time</b>       | 95th percentile under 300 ms for reads and 600 ms for writes, excluding payment provider latency.                                                                           |
| <b>NFR-03</b> | <b>Capacity</b>            | 10,000 active registries and 2,000 contributions per hour at peak, scalable tenfold without redesign. Keyset pagination on all lists; reports served from daily aggregates. |
| <b>NFR-04</b> | <b>Availability</b>        | 99.5% monthly for guest pages and checkout. Minimum two replicas per service; zero-downtime deployments.                                                                    |
| <b>NFR-05</b> | <b>Financial integrity</b> | Each contribution credited once, each payout executed at most once, balanced ledger, hourly reconciliation with the PayHero wallet.                                         |
| <b>NFR-06</b> | <b>Security</b>            | OWASP ASVS Level 2, TLS throughout, encrypted secrets, MFA for staff, audit trail on all financial and configuration changes.                                               |
| <b>NFR-07</b> | <b>Privacy</b>             | Data minimisation, consent records, data subject requests within statutory timelines, documented hosting location and transfer safeguards.                                  |
| <b>NFR-08</b> | <b>Accessibility</b>       | WCAG 2.2 Level AA.                                                                                                                                                          |
| <b>NFR-09</b> | <b>Localisation</b>        | English at launch. Text externalised for Kiswahili.                                                                                                                         |
| <b>NFR-10</b> | <b>Recovery</b>            | Daily encrypted backups, point-in-time recovery. RPO 24 hours, RTO 4 hours.                                                                                                 |
| <b>NFR-11</b> | <b>Observability</b>       | Structured logs, distributed tracing, alerts on failed payouts, reconciliation variances, error rates and latency.                                                          |
| <b>NFR-12</b> | <b>Maintainability</b>     | Codevertex platform stack and conventions, automated tests, documented APIs.                                                                                                |

## SECTION 5

# Solution architecture

**Figure 1.** Solution architecture

## 5.1 Components

| Component         | Status             | Function                                                                                                                                                                                              |
|-------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hadia-ui          | New                | Next.js progressive web app: website, guest pages, owner dashboard, administration console. Uses the Codevertex shared UI library for payments, sign-in, data tables, legal links and cookie consent. |
| hadia-api         | New                | Go service for registries, items, reservations, premium purchases, gift log, statistics and consent records. Does not move money.                                                                     |
| auth-api          | Existing, extended | Tenant, staff single sign-on with MFA, roles, terms acceptance, sessions. Extended with passwordless mobile sign-in for consumers.                                                                    |
| treasury-api      | Existing, extended | Ledger, PayHero gateway, checkout, escrow, approvals, payout dispatch, reconciliation, statements, eTIMS invoicing. Extended with scheduled commission sweep and per-registry contribution limits.    |
| treasury-ui       | Existing           | Finance console: ledger, escrow accounts, approvals, wallet, reports.                                                                                                                                 |
| notifications-api | Existing           | SMS (Africa's Talking), WhatsApp, email (Brevo), push. Hadia message templates added.                                                                                                                 |
| subscriptions-api | Existing           | Hadia's platform plan and feature entitlements.                                                                                                                                                       |

---

|                     |                 |                                                                                                                                                              |
|---------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shared<br>libraries | <b>Existing</b> | Authentication client, service client with retries and circuit breakers, event outbox, cache and job leases, pagination, HTTP middleware, shared UI library. |
|---------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|

---

## 5.2 Architecture principles

- **Single data owner.** hadia-api references treasury records by ID. Balances are read from treasury.
- **Single payment path.** Only treasury-api communicates with PayHero.
- **Event driven.** Treasury publishes escrow events through a transactional outbox to NATS. hadia-api consumes them idempotently.
- **Replica safe.** Scheduled jobs run once per period across all replicas using shared leases.
- **Standard stack.** Go with Ent and Atlas migrations, Next.js, Helm and ArgoCD.

## SECTION 6

# Funds management and payouts

## 6.1 Custody

Hadia operates as a tenant on the Codevertex platform with its own PayHero Team account, registered in Hadia's name and verified to PayHero KYC tier 3. The Team wallet is isolated and used only by Hadia. All contributions are received into this wallet. Treasury maintains a separate sub-ledger for each registry, so the wallet balance is fully allocated between registries and Hadia's earned commission at all times.

Codevertex does not receive or hold Hadia funds or customer funds.

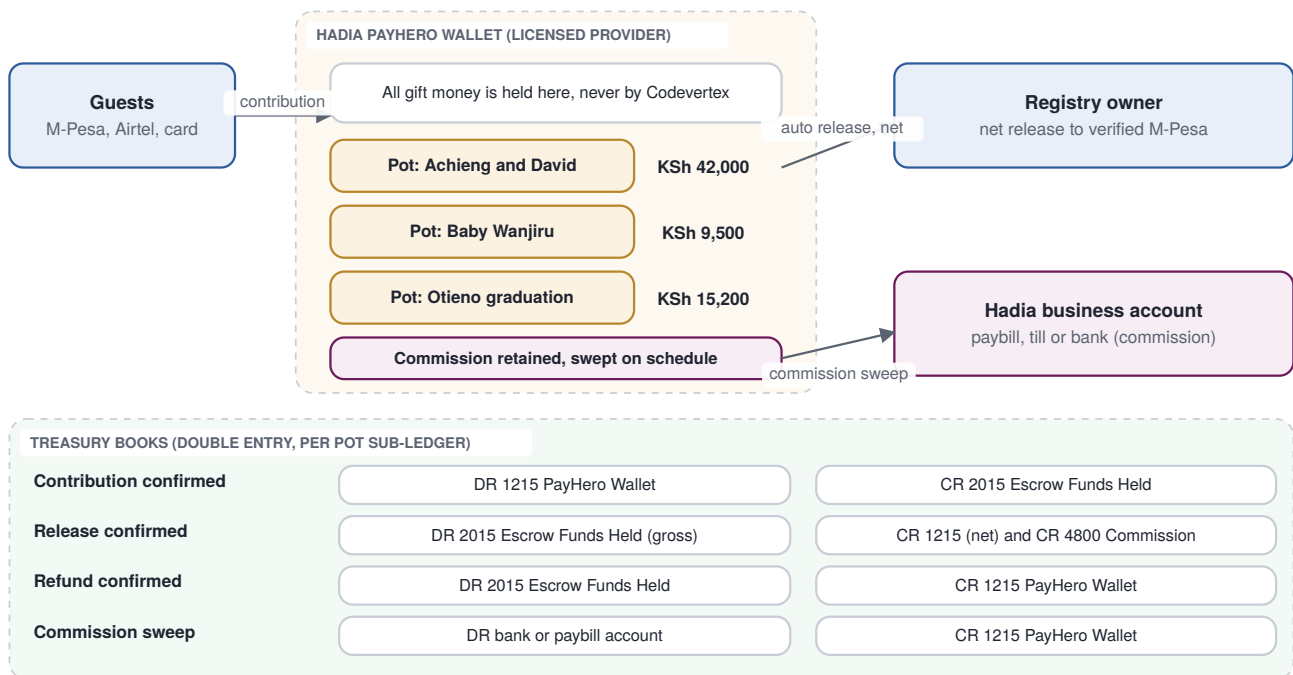


Figure 2. Funds flow and accounting entries

## 6.2 Registry escrow account

Each published registry has an escrow account in treasury. The account holds the registry title, the owner as beneficiary with a verified M-Pesa number, the target, payout rules and gift items. The registry ID is used as the external reference, which prevents duplicate accounts on retry.

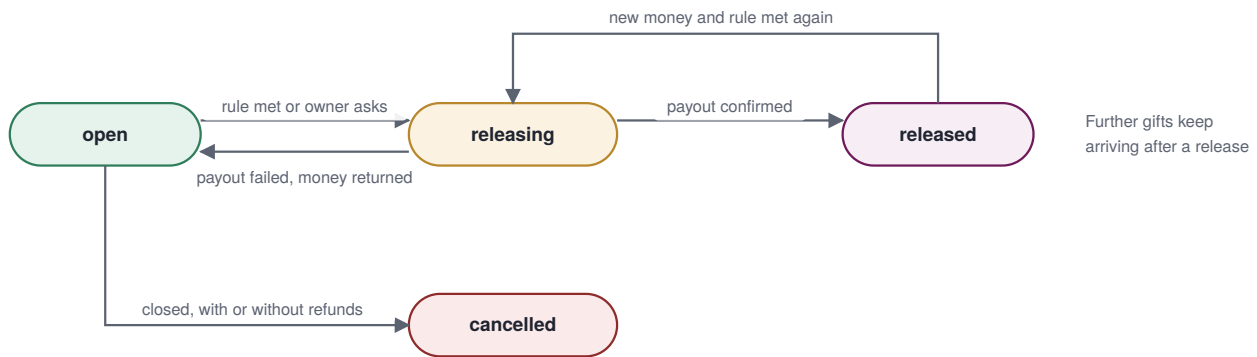


Figure 3. Escrow account lifecycle

### 6.3 Payout configuration

| Setting             | Level                         | Default                                             | Behaviour                                                                                        |
|---------------------|-------------------------------|-----------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Automatic payout    | Registry                      | On                                                  | Payouts run when a rule is met, checked every 5 minutes                                          |
| Payout threshold    | Registry                      | KES 10,000 standard; every contribution for premium | Full balance paid when the threshold is reached                                                  |
| Hold until          | Registry                      | Event date plus 2 days                              | Remaining balance paid after this date                                                           |
| Approval limit      | Hadia                         | KES 150,000 per payout                              | Payouts at or above the limit require finance approval with OTP, bound to amount and destination |
| Commission          | Hadia, with registry override | 4%                                                  | Deducted at payout and recorded as Hadia income                                                  |
| Contribution limits | Registry                      | KES 50 minimum, KES 150,000 maximum                 | Contributions outside the range are rejected before the STK push                                 |
| Maximum balance     | Platform                      | Agreed with Hadia                                   | Contributions exceeding the cap are declined                                                     |
| Maximum hold        | Platform                      | 180 days                                            | Funds cannot be held beyond this period                                                          |
| Commission sweep    | Hadia                         | Daily above KES 5,000, KES 2,000 reserve            | Earned commission transferred to Hadia's business account                                        |

## 6.4 Payout process

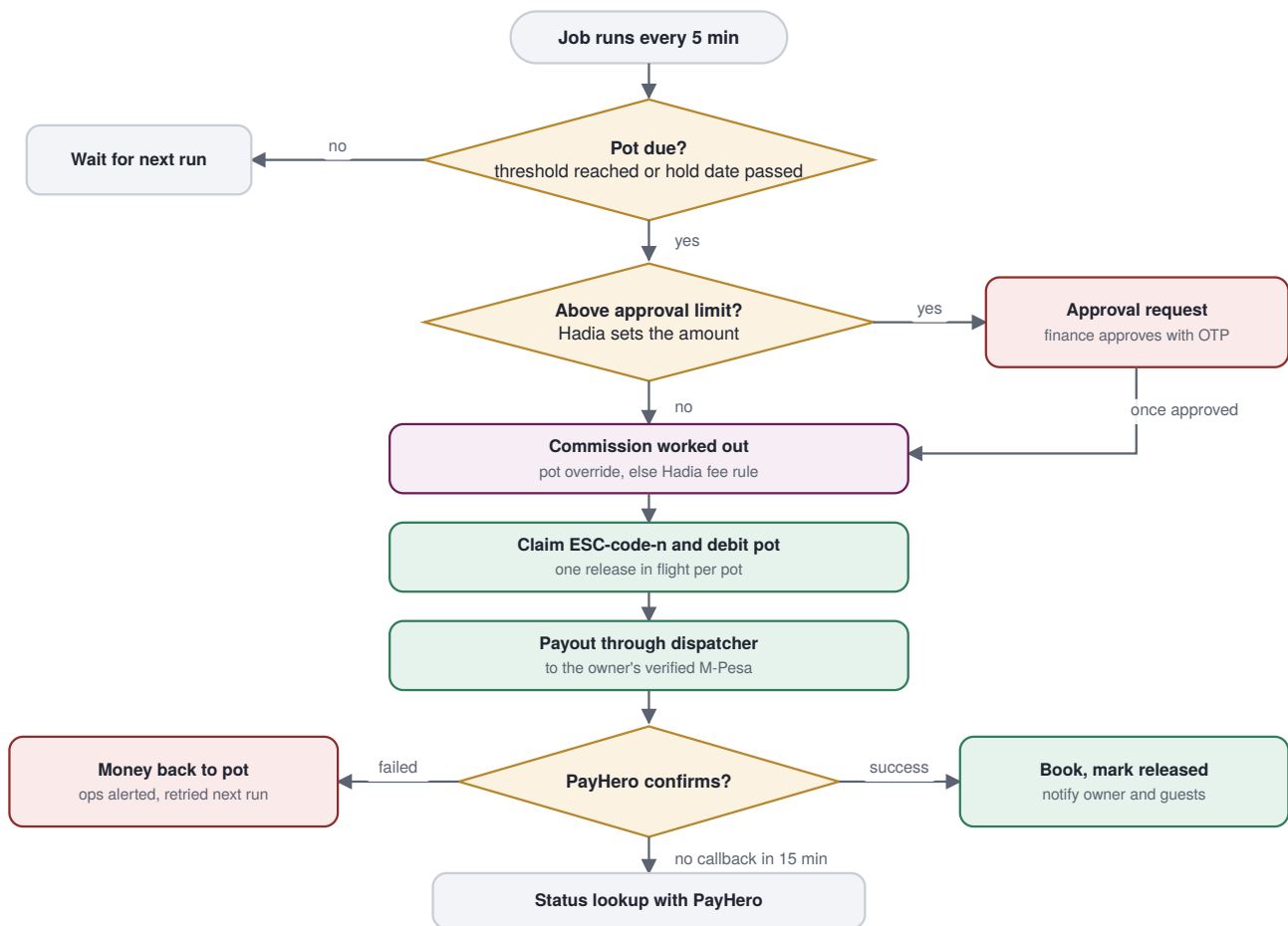


Figure 4. Automated payout process

Each payout attempt is registered under a unique reference before PayHero is called. Only one payout may be active per registry. A failed payout returns the funds to the registry balance. Payouts without a callback after 15 minutes are confirmed by status enquiry. All results are verified with PayHero before posting.

## 6.5 Commission

Commission is deducted from the gross payout and the net amount is paid to the owner. Commission is recorded in income account 4800 and retained in the Team wallet until swept.

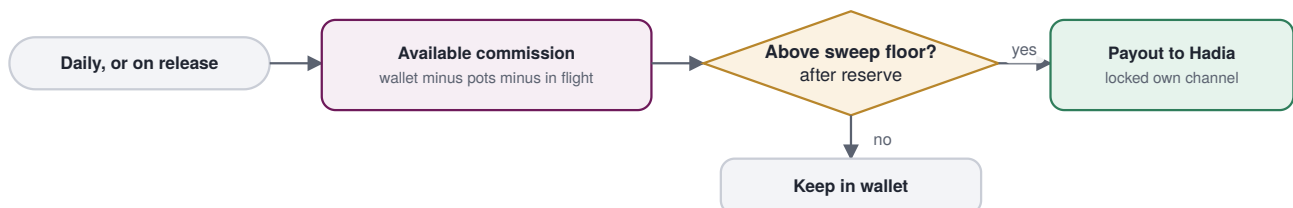


Figure 5. Commission sweep

1. The sweep runs daily, or after each payout if configured.
2. Available commission equals the wallet balance less registry balances, payouts in progress and the operating reserve.
3. Amounts above the sweep minimum are paid to Hadia's verified business channel. Changing the destination requires MFA and re-verification.
4. The transfer is posted from the PayHero wallet account to Hadia's bank or paybill account. The sweep is suspended if reconciliation reports a shortfall.

Manual withdrawal of commission remains available as an alternative.

## 6.6 Reconciliation

Treasury compares the total of registry balances and payouts in progress with the PayHero wallet balance every hour. Any shortfall is raised in the exception queue and suspends the commission sweep until resolved. Each registry statement is available in PDF, CSV and Excel.

## SECTION 7

# Process workflows

Each workflow defines its trigger, steps, automation and exception handling. Steps marked **AUTO** run without user or staff action.

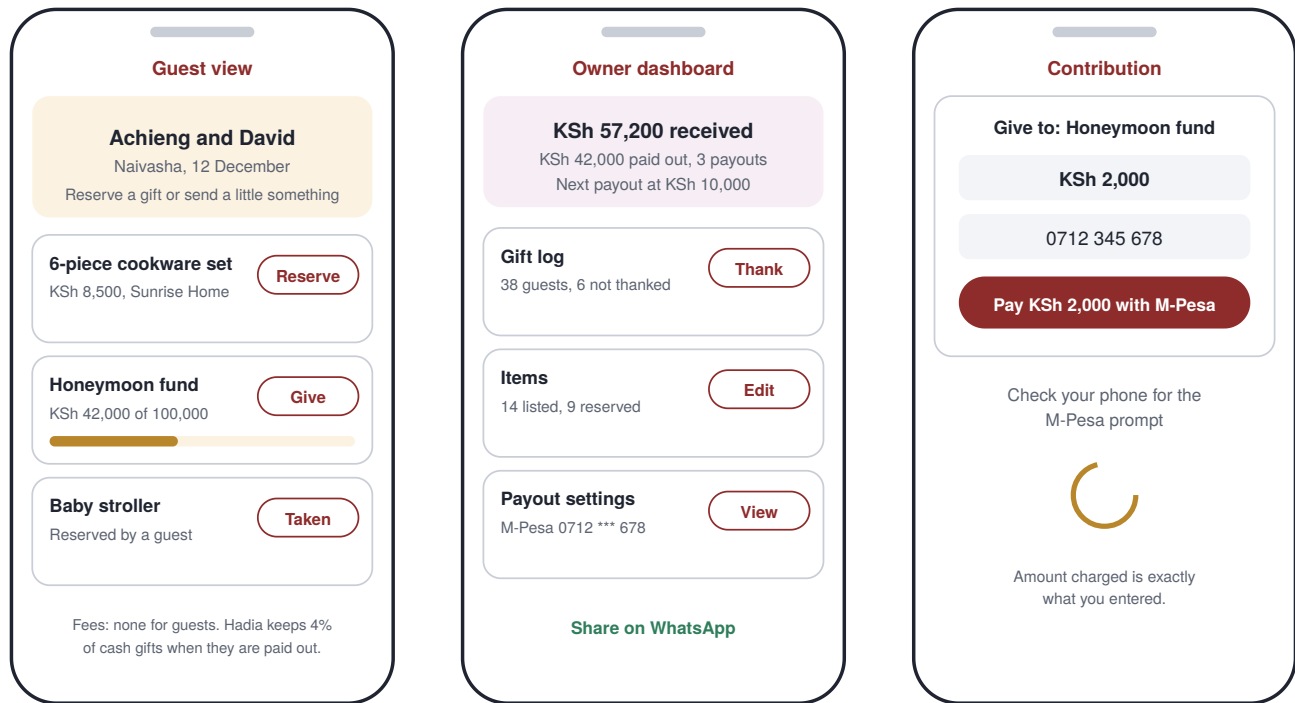


Figure 6. Key screens (low fidelity)

## 7.1 Hadia business onboarding



Figure 7. Hadia business onboarding

| Step | Action                                                                                                                   | Responsible    |
|------|--------------------------------------------------------------------------------------------------------------------------|----------------|
| 1    | Submit company documents: certificate of incorporation, CR12, KRA PIN, director identification, bank or paybill details. | Hadia          |
| 2    | Create the Hadia tenant and assign the platform plan with escrow and M-Pesa features.                                    | Codevertex     |
| 3    | Create or link the Hadia PayHero Team and complete KYC to tier 3.                                                        | Hadia, PayHero |
| 4    | Synchronise the paybill or till and map the wallet and channels to the chart of accounts.                                | Codevertex     |
| 5    | Accept the escrow terms in treasury.                                                                                     | Hadia          |
| 6    | Configure commission, payout policy, approval limit, contribution limits and sweep schedule.                             | Hadia finance  |

- |   |                                                                                          |                      |
|---|------------------------------------------------------------------------------------------|----------------------|
| 7 | Execute live KES 1 tests for contribution, payout, refund and sweep, and reconcile each. | Codevertex,<br>Hadia |
|---|------------------------------------------------------------------------------------------|----------------------|

## 7.2 Registry owner onboarding

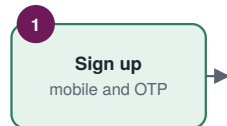


Figure 8. Registry owner onboarding

| Step | Action                               | System behaviour                                                                                                             |
|------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| 1    | Owner enters a mobile number         | One-time password sent by SMS, WhatsApp fallback. Codes hashed, 5-minute expiry, rate limited per number and IP. <b>AUTO</b> |
| 2    | Owner confirms age and accepts terms | Version and timestamp recorded                                                                                               |
| 3    | Owner creates the registry           | Type, title, date, location, cover photo, description                                                                        |
| 4    | Owner verifies the payout number     | One-time password sent to the payout number                                                                                  |
| 5    | Registry published                   | Escrow account opened in treasury with default rules <b>AUTO</b>                                                             |
| 6    | Owner shares the registry            | Short link and preview image generated <b>AUTO</b>                                                                           |

**Exception:** if the escrow account cannot be opened, the registry is saved as a draft and the request is retried in the background. Item entry continues uninterrupted.

## 7.3 Item listing

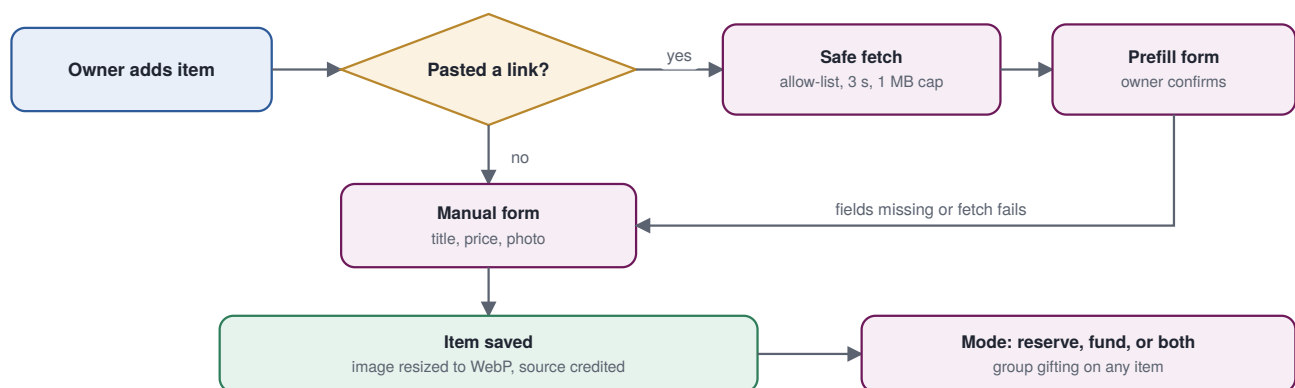


Figure 9. Item listing

1. The owner pastes a product link or selects manual entry.
2. For links, the server retrieves public page metadata under strict controls: public addresses only, 3-second timeout, 1 MB limit, no cookies.
3. Title, image and price are prefilled for owner confirmation. Where no data is found, the manual form is shown.
4. Photos are converted to WebP at 640 and 1280 pixels, stored in object storage and served through the CDN. The source store is credited.
5. The owner sets the item mode: reserve, fund, or both.

## 7.4 Sharing

1. Each registry receives a short link; premium registries may set a custom link.
2. A 1200 by 630 pixel preview image under 300 KB is generated for WhatsApp and social platforms. **AUTO**
3. Sharing options: WhatsApp, SMS, copy link and QR code.

## 7.5 Item reservation

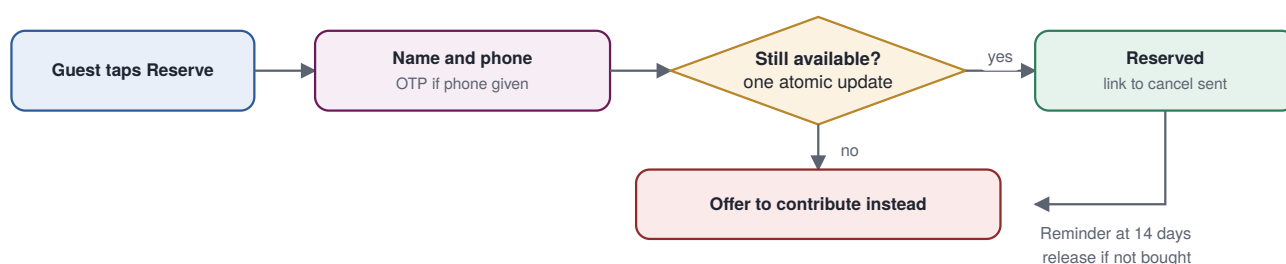


Figure 10. Item reservation

1. The guest selects Reserve and enters a name and, optionally, a mobile number.
2. The reservation is recorded in a single conditional update. Concurrent requests for the last unit result in one reservation; other guests are offered a contribution instead.
3. The guest receives a cancellation link. The item is shown as reserved to all guests.
4. After 14 days without purchase confirmation, reminders are sent and the owner may release the item. **AUTO**

## 7.6 Cash contribution

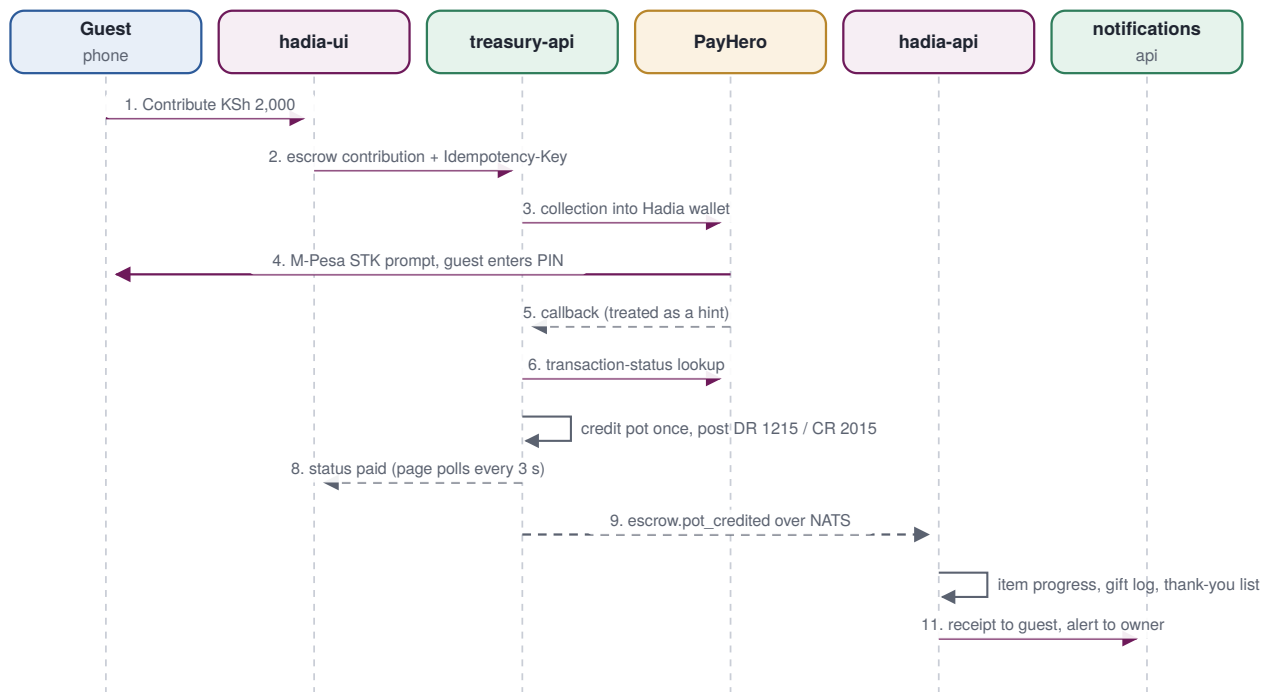


Figure 11. Cash contribution sequence

1. The guest selects an item or fund and enters an amount and mobile number. Limits are validated.
2. hadia-ui submits the contribution to treasury with an idempotency key.
3. Treasury initiates a PayHero collection into the Hadia wallet. The guest authorises the STK push.
4. On callback, treasury verifies the transaction with PayHero, credits the registry once and posts the ledger entry. **AUTO**
5. The checkout page confirms payment. Treasury publishes the credit event.
6. hadia-api updates item progress and the gift log and triggers the guest receipt and owner notification. **AUTO**

**Exceptions:** failed or cancelled prompts leave the contribution pending and the paybill option is offered. Repeated failures to one number (five in 24 hours) suspend further prompts to that number. Missing callbacks are resolved by status enquiry.

## 7.7 Registry management

1. The dashboard shows amounts received, paid out, pending and the next payout.
2. Owners may change the target, threshold (within Hadia's limits) and event date. The payout number changes only after verification of the new number and never during an active payout.
3. Contributions above an item's price remain in the registry balance and are paid out with it.
4. After the hold period the registry closes to new contributions. Owners may reopen it.

## 7.8 Automated payout

Registries meeting a payout rule are paid every 5 minutes to the verified number, net of commission, with approval required only at or above Hadia's limit. The owner receives an SMS showing gross amount, commission and net amount. **AUTO** Full process in [Funds management and payouts](#).

## 7.9 Commission and sweep

Commission is deducted at payout, recorded as Hadia income and transferred daily to Hadia's business account above the sweep minimum. **AUTO** eTIMS invoices for Hadia's fees are issued through treasury's KRA integration.

## 7.10 Premium upgrade

1. The owner selects Upgrade and is shown the features and price.
2. hadia-api creates a payment intent in treasury and opens the shared payment modal.
3. On confirmation, treasury records the sale as Hadia revenue and hadia-api enables the custom link, printable list and instant payouts. **AUTO**
4. A receipt is issued. Refunds follow the refund policy.

## 7.11 Cancellation and refunds

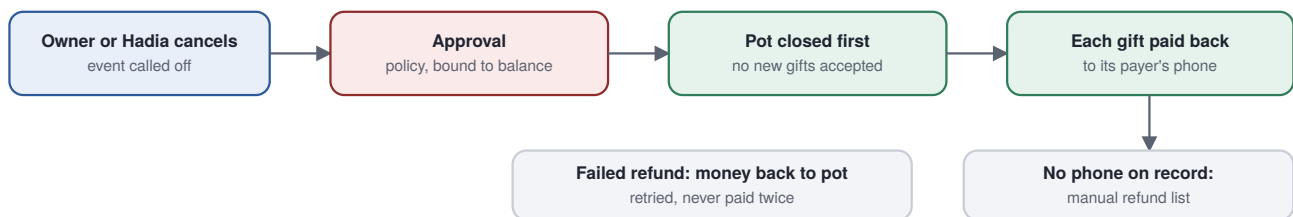


Figure 12. Cancellation and refunds

1. The owner requests cancellation, or Hadia cancels following a fraud or policy finding.
2. The refund is authorised under the approval policy, bound to the registry and its balance.
3. The registry is closed to new contributions. Each contribution is refunded to the paying number under its own reference. **AUTO**
4. Failed refunds return to the registry balance for retry. Repeated cancellation requests never duplicate a refund. Contributions without a number are listed for manual settlement.

## 7.12 Exceptions and reconciliation

1. Reconciliation runs hourly. **AUTO**
2. The exception queue lists failed payouts, name mismatches, manual refunds, reconciliation variances, reported registries and pending approvals.
3. Each entry records the facts, recommended action and resolver. Finance receives a daily summary.

## 7.13 Thank-you tracking

1. All contributions and reservations appear in the gift log with guest messages.
2. The owner sends a prepared WhatsApp thank-you and marks the guest as thanked.
3. A reminder listing outstanding thank-yous is sent one week after the event. Owners may disable it.

## 7.14 Data subject requests

1. An owner or guest submits a request for access, correction or deletion.
2. Identity is confirmed by one-time password and the statutory deadline is recorded.
3. Access requests produce a machine-readable export.
4. Deletion erases or anonymises personal data. Financial records required by law are retained for the statutory period under a reference, and the requester is informed.

## SECTION 8

# Data design

hadia-api owns the registry domain and references other services by ID: the owner's user ID in auth-api and the escrow account and payment intent IDs in treasury.

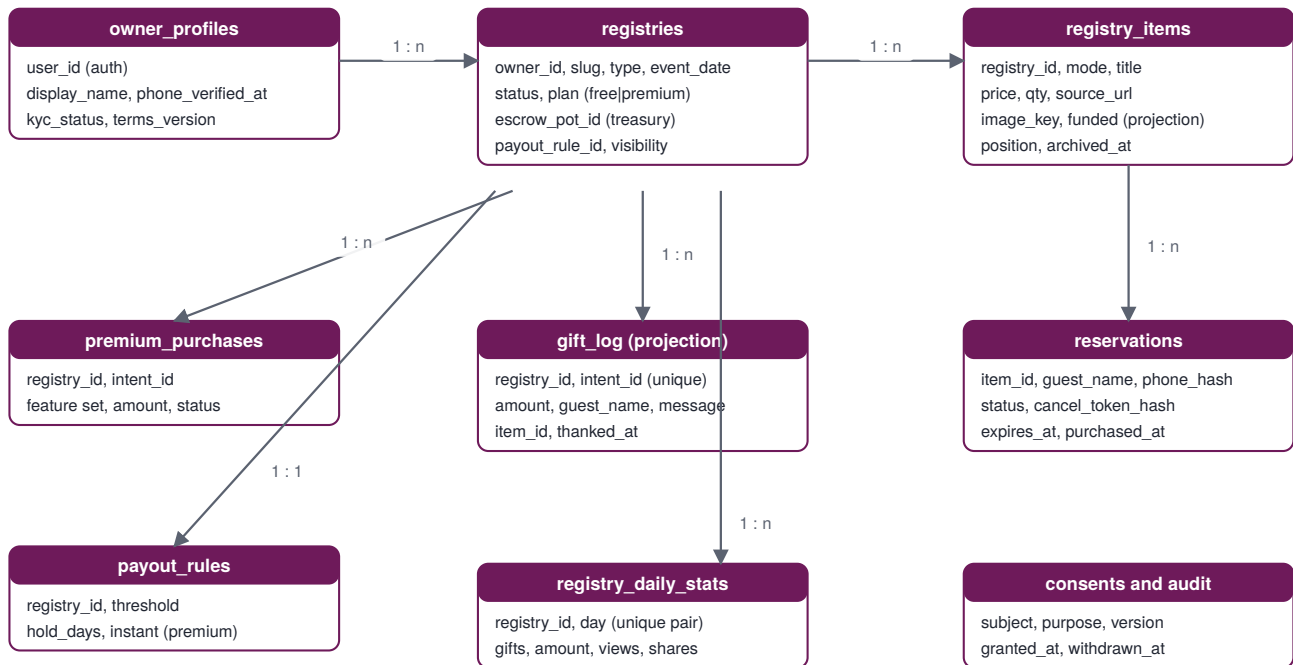


Figure 13. hadia-api data model

## 8.1 Scalability measures

| Measure                                                                                    | Purpose                                                   |
|--------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Tenant ID and query-specific indexes on every table, managed by versioned Atlas migrations | Consistent query performance and controlled schema change |
| Keyspace pagination on all lists                                                           | Constant response time regardless of page depth           |
| Unique constraint on payment intent ID in the gift log                                     | No duplicate records from redelivered events              |
| Counters updated in a single SQL statement                                                 | Correct totals under concurrency                          |
| Daily aggregate table for reporting                                                        | Reports independent of raw data volume                    |
| Guest numbers hashed for matching and encrypted where display is required                  | Data protection                                           |
| Images in object storage; keys only in the database                                        | Small, fast database                                      |

## SECTION 9

# Interfaces, events and notifications

## 9.1 hadia-api endpoints

| Area           | Endpoints                                                                                             | Access               |
|----------------|-------------------------------------------------------------------------------------------------------|----------------------|
| Registries     | GET, POST /registries; GET, PATCH /registries/{id}; POST /registries/{id}/publish, /close, /cancel    | Owner                |
| Items          | GET, POST /registries/{id}/items; PATCH, DELETE /items/{id}; POST /items/preview; POST /items/reorder | Owner                |
| Public         | GET /r/{code}; POST /r/{code}/items/{item}/reserve; POST /reservations/{token}/cancel                 | Public, rate limited |
| Gifts          | GET /registries/{id}/gifts; POST /gifts/{id}/thanked; GET /registries/{id}/statement                  | Owner                |
| Premium        | POST /registries/{id}/premium                                                                         | Owner                |
| Privacy        | POST /privacy/export, /privacy/delete; GET /privacy/requests                                          | Owner; guest by OTP  |
| Administration | Search, moderation, exceptions, settings, reports                                                     | Staff by role        |

## 9.2 Treasury endpoints

| Function                            | Endpoint                                                                       |
|-------------------------------------|--------------------------------------------------------------------------------|
| Escrow account create, read, update | POST, GET, PUT /{tenant}/escrow/pots                                           |
| Guest contribution                  | POST /api/v1/pay/{tenant}/escrow/{code}/contributions                          |
| Public account view                 | GET /api/v1/pay/{tenant}/escrow/{code}                                         |
| Payout; cancellation with refunds   | POST /{tenant}/escrow/pots/{pot}/release; /cancel?refund=true                  |
| Commission rule, readiness, summary | GET, PUT /{tenant}/escrow/fee-rule; GET /escrow/readiness; GET /escrow/summary |
| Statements                          | GET /{tenant}/escrow/pots/{pot}/statement; /statement/document                 |
| Premium sale                        | POST /{tenant}/payments/intents                                                |

## 9.3 Events

| Event                  | Publisher | hadia-api action                                                            |
|------------------------|-----------|-----------------------------------------------------------------------------|
| escrow.pot_created     | treasury  | Mark escrow ready; publish registry                                         |
| escrow.pot_credited    | treasury  | Update gift log, item funding and statistics; send receipt and notification |
| escrow.dispatched      | treasury  | Record payout; notify owner of gross, commission and net                    |
| escrow.dispatch_failed | treasury  | Raise exception; notify owner of retry                                      |
| escrow.cancelled       | treasury  | Close registry; notify guests of refunds                                    |

|                                                        |           |                                         |
|--------------------------------------------------------|-----------|-----------------------------------------|
| payment.succeeded                                      | treasury  | Activate premium features               |
| hadia.registry_published,<br>hadia.reservation_created | hadia-api | Consumed by notifications and analytics |

Consumers are idempotent on event ID. Events are retained in the outbox during broker outages.

## 9.4 Notifications

| Message                             | Recipient       | Channel                              |
|-------------------------------------|-----------------|--------------------------------------|
| One-time password                   | Owner, guest    | SMS; WhatsApp fallback               |
| Contribution receipt                | Guest           | SMS or WhatsApp                      |
| New contribution or reservation     | Owner           | WhatsApp or SMS; daily digest option |
| Payout completed or delayed         | Owner           | SMS                                  |
| Reservation and thank-you reminders | Guest, owner    | WhatsApp or SMS                      |
| Refund completed                    | Guest           | SMS                                  |
| Exception summary, approval request | Hadia finance   | Email, WhatsApp                      |
| Product updates                     | Opted-in owners | Email or WhatsApp with unsubscribe   |

## SECTION 10

# Reporting and analytics

| Report                                        | Audience         | Source                                              |
|-----------------------------------------------|------------------|-----------------------------------------------------|
| Registry dashboard                            | Owner            | Treasury balances and payouts; gift log             |
| Contributions and value by period             | Hadia            | Daily aggregates                                    |
| Active registries, new registries, conversion | Hadia            | Daily aggregates and page view counters             |
| Commission and premium revenue                | Hadia finance    | Treasury ledger, trial balance, income statement    |
| Funds held, in progress and available         | Hadia finance    | Treasury escrow summary                             |
| Payout performance                            | Hadia operations | Payout records: time to completion, failure reasons |
| Registry statement                            | Owner, finance   | Treasury statement in PDF, CSV, Excel               |

Analytical queries run on aggregate tables and may use the read replica. Dashboard results are cached for 60 seconds and invalidated by the related event.

## SECTION 11

# Security

| Threat                             | Controls                                                                                                                                                   |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Account takeover                   | OTP rate limits by number and IP, hashed codes, short expiry, session revocation, MFA for staff, re-verification for payout changes                        |
| Payout redirection                 | Payout number verified by OTP, changes blocked during active payouts, approvals bound to amount and destination, name check on first payout, change alerts |
| Duplicate charges or payouts       | Idempotency keys, unique payout references, single settlement path, provider verification of callbacks                                                     |
| Fraudulent registries              | Moderation, guest reporting, registry creation limits, balance caps, first-payout hold for new owners                                                      |
| Money laundering                   | Contribution and balance limits, KYC, verified payout numbers, pattern monitoring, escalation to Hadia's compliance contact                                |
| Server-side request forgery        | Public address allow-list with post-resolution DNS checks, redirect validation, time and size limits                                                       |
| Injection and cross-site scripting | Parameterised queries, output encoding, Content Security Policy, sanitised input                                                                           |
| Automated abuse                    | Rate limiting, STK prompt guard, Cloudflare WAF                                                                                                            |
| Data exposure                      | TLS, encryption at rest, least-privilege roles, payout details excluded from public pages, staff access logging                                            |
| Card data                          | Provider-hosted card checkout; no card data on Hadia systems                                                                                               |

## SECTION 12

# Legal and regulatory compliance

This chapter sets out the legal obligations applicable to Hadia and the controls that address them. It does not constitute legal advice. Items marked **Confirm** require an opinion from Hadia's advocate before launch.

## 12.1 Roles and responsibilities

| Party                                    | Role                                                                                                     | Obligations                                                                                                                          |
|------------------------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Hadia                                    | Service operator and data controller; holds customer funds for beneficiaries through its PayHero account | User terms, fees, payouts, complaints, ODPC registration, tax, KYC and AML cooperation                                               |
| Codevertex Africa Limited                | Software provider and data processor                                                                     | Secure development and operation, processing on Hadia's instructions under a data processing agreement, breach notification to Hadia |
| PayHero                                  | Payment service provider                                                                                 | Collections, wallet custody, payouts, KYC and AML, card processing                                                                   |
| Africa's Talking, Meta (WhatsApp), Brevo | Sub-processors                                                                                           | Message delivery under their terms                                                                                                   |

## 12.2 Applicable law

| Law                                                                                | Obligation                                                                                                                                                                                        | Control                                                                                                                                                                             |
|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Constitution of Kenya, Article 31; Data Protection Act, 2019 and Regulations, 2021 | Lawful and minimal processing, ODPC registration, DPIA, data subject rights, 72-hour breach notification, transfer safeguards, consent for direct marketing, parental consent for children's data | Privacy notice, consent records, data minimisation, DPIA, ODPC registration, data request workflow, breach procedure, documented hosting location, opt-in marketing                 |
| Consumer Protection Act, 2012                                                      | No misleading representations; for internet agreements, disclosure before acceptance, express acceptance, error correction and a copy of the agreement                                            | Fees and payout timing disclosed before sign-up and payment, versioned click-through acceptance, downloadable terms, verified marketing claims                                      |
| Competition Act, 2010                                                              | No false or misleading representations, including reviews and testimonials                                                                                                                        | Consented testimonials only; claims supported by records                                                                                                                            |
| National Payment System Act, 2011 and Regulations                                  | Payment services and holding funds for others are regulated by the Central Bank of Kenya                                                                                                          | Funds held and paid by PayHero; capped hold periods and balances; per-registry ledgers; hourly reconciliation. <b>Confirm</b> whether Hadia requires CBK authorisation or guidance. |
| Proceeds of Crime and Anti-Money Laundering Act, 2009                              | Prevent laundering; maintain records; cooperate with reporting institutions                                                                                                                       | KYC tier 3, verified payout numbers, contribution and balance caps, transaction monitoring, record retention, designated compliance contact                                         |

|                                                                                 |                                                                                                    |                                                                                                                                                        |
|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Computer Misuse and Cybercrimes Act, 2018                                       | System and data protection; handling of unlawful content                                           | Security controls, audit logging, moderation and reporting, incident response                                                                          |
| Kenya Information and Communications Act; Consumer Protection Regulations, 2010 | Validity of electronic records; no unsolicited messages; opt-out                                   | Logged acceptance, transactional messaging by default, STOP and unsubscribe on marketing, registered sender ID                                         |
| Tax Procedures Act, 2015; VAT Act, 2013; eTIMS                                  | Record retention of at least five years; electronic tax invoices; VAT registration above threshold | Ledger in treasury, gifts recorded as liabilities, commission and premium as income, eTIMS invoicing. <b>Confirm</b> VAT and excise treatment of fees. |
| Children Act, 2022; Data Protection Act, section 33                             | Parental consent and best interests for children's data                                            | Owners aged 18 or over, parental confirmation before child names or photos, unlisted registries by default                                             |
| Persons with Disabilities Act; Constitution, Article 54                         | Accessible public services                                                                         | WCAG 2.2 AA                                                                                                                                            |
| Copyright Act, 2001; Trade Marks Act                                            | Respect for third-party content and marks                                                          | Licensed fonts and images, credited product thumbnails with takedown, descriptive use of store names only                                              |
| Business Registration Service Act; Companies Act, 2015                          | Trade under registered name with business identification                                           | Registered name, number, address and contacts on every page and receipt                                                                                |
| EU and UK GDPR; US CAN-SPAM and COPPA (diaspora users)                          | Equivalent privacy rights, email unsubscribe, no data from children under 13                       | Kenyan controls applied, prior consent for optional cookies, unsubscribe in every email, service not directed at children                              |

## 12.3 Compliance checklist

| # | Requirement       | Implementation                                                                                                                          | Owner                               |
|---|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| 1 | Privacy policy    | Controller, processors, purposes, lawful bases, retention, rights, transfers and contacts. Linked from every page, form and message.    | Codevertex draft; advocate approval |
| 2 | Terms of service  | Owner, guest and escrow terms; acceptable use; fees; liability; disputes; Kenyan law. Versioned acceptance; 14 days' notice of changes. | Codevertex draft; advocate approval |
| 3 | Refund policy     | Refund grounds, timelines, premium refunds within 14 days if unused, fee treatment. Enforced by the refund workflow.                    | Hadia                               |
| 4 | Cookie policy     | All cookies listed by category, purpose and duration. Strictly necessary cookies only at launch.                                        | Codevertex                          |
| 5 | Cookie consent    | No pre-selected options; equal reject and accept controls; settings available from the footer.                                          | Codevertex                          |
| 6 | Form consents     | Purpose stated for each field; separate unticked consents; consent versioned and withdrawable.                                          | Codevertex                          |
| 7 | Data minimisation | Guests provide name and number only. No identity numbers, dates of birth or location tracking.                                          | Codevertex                          |

|    |                           |                                                                                                                |            |
|----|---------------------------|----------------------------------------------------------------------------------------------------------------|------------|
| 8  | Third-party audit         | No advertising trackers. Register of third parties and data shared. Analytics only with consent.               | Codevertex |
| 9  | No dark patterns          | No pre-ticked boxes, false urgency or confirm-shaming. Cancellation and deletion as simple as sign-up.         | Codevertex |
| 10 | No hidden fees            | Guests pay the amount entered. Commission and payout terms shown before registry creation and on every payout. | Codevertex |
| 11 | Genuine reviews           | Testimonials published only with written consent, retained on file.                                            | Hadia      |
| 12 | Substantiated claims      | Marketing statements reflect actual funds handling and payout terms.                                           | Hadia      |
| 13 | Alternative text          | Required for uploaded photos; descriptive text on all content images.                                          | Codevertex |
| 14 | Colour contrast           | 4.5:1 for text and 3:1 for controls.                                                                           | Codevertex |
| 15 | Keyboard navigation       | Full keyboard operation, visible focus, managed dialogs, skip link, automated accessibility testing.           | Codevertex |
| 16 | Business details          | Registered name, number, address and contacts in the footer and on receipts.                                   | Codevertex |
| 17 | Children's data consent   | Owner age confirmation and parental consent before child information.                                          | Codevertex |
| 18 | Unsubscribe               | One-click unsubscribe in email; STOP in SMS and WhatsApp marketing.                                            | Codevertex |
| 19 | Licensed fonts and images | Open-licence fonts and licensed imagery, recorded in a licence register.                                       | Codevertex |
| 20 | Data deletion             | Self-service export and deletion with deadline tracking and statutory retention.                               | Codevertex |

## 12.4 Website content review

| Current text                                   | Issue                                                  | Recommendation                                                                             |
|------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------|
| "Loved by 400+ families"                       | Unverified figure                                      | Remove until supported by platform data                                                    |
| Achieng and David testimonial                  | Not a verified customer statement                      | Replace with consented pilot testimonials                                                  |
| "Funds go straight to you"                     | Funds are held and paid out by rule, net of commission | "Cash gifts are held securely and paid to your M-Pesa automatically, less Hadia's 4% fee." |
| "Your money, kept safe"                        | Requires substantiation                                | Link to a page describing custody, ledgers and reconciliation                              |
| "You only ever share in what's actually given" | Incomplete fee disclosure                              | "Free to create. 4% of cash gifts. Optional premium at KES 300 per event."                 |
| "Carrefour, Naivas, TikTok Shop"               | Third-party marks                                      | Text references only; no logos or implied partnership                                      |

## 12.5 Record retention

| Record                                            | Retention                                                         | Basis                                  |
|---------------------------------------------------|-------------------------------------------------------------------|----------------------------------------|
| Contributions, payouts, refunds, invoices, ledger | 7 years                                                           | Tax Procedures Act; AML record keeping |
| Owner accounts and registries                     | Until deletion, or 24 months after last activity, then anonymised | Data minimisation                      |
| Reservation contact details                       | 12 months after the event                                         | Reservation and thank-you management   |
| One-time passwords                                | 5 minutes, hashed                                                 | Security                               |
| Consent and terms acceptance                      | Account life plus 5 years                                         | Evidence of consent                    |
| Security and audit logs                           | 12 months                                                         | Incident investigation                 |

## 12.6 Legal deliverables

- Privacy notice, cookie policy, terms of service, refund policy, acceptable use policy
- Data processing agreement and sub-processor list
- Data protection impact assessment and record of processing activities
- Breach response procedure and data request procedure
- Third-party and licence registers, accessibility statement

## SECTION 13

# Project delivery

Delivery follows Agile Scrum with weekly demonstrations on the staging environment.

## 13.1 Methodology

| Practice           | Approach                                                                                               |
|--------------------|--------------------------------------------------------------------------------------------------------|
| Iterations         | One-week Sprint 0, four two-week sprints, one-week Sprint 5, hardening week, launch week               |
| Demonstrations     | Weekly 30-minute review with the product owner; decisions logged in the backlog                        |
| Backlog            | User stories traced to FR and NFR identifiers, with acceptance criteria                                |
| Definition of done | Code reviewed, tests and accessibility checks passed, documented, deployed to staging, accepted        |
| Environments       | Development, staging and production on the Codevertex platform                                         |
| Quality assurance  | Unit, integration, end-to-end and contract tests; load testing before launch                           |
| Release management | CI on GitHub Actions; GitOps deployment through ArgoCD                                                 |
| Reporting          | Weekly written status: progress, next steps, risks, decisions required                                 |
| Change control     | Equal-size scope substitutions within the fixed price; other changes estimated and approved in writing |

## 13.2 Project team

| Role                               | Allocation                                  |
|------------------------------------|---------------------------------------------|
| Project lead and business analyst  | Part time, weeks 1 to 12                    |
| Full-stack engineers (Go, Next.js) | Two, weeks 1 to 12                          |
| Product designer                   | Part time, weeks 1 to 5; reviews thereafter |
| QA engineer                        | Part time, weeks 3 to 12                    |
| Platform and DevOps engineer       | As required                                 |

## 13.3 Implementation schedule

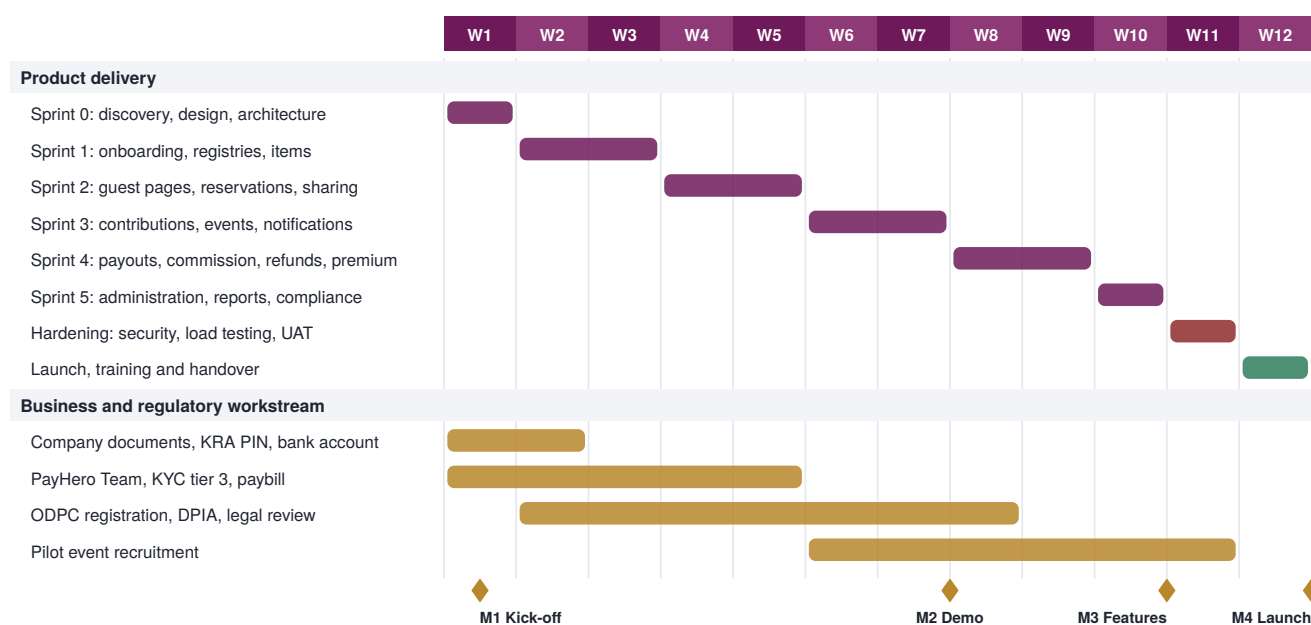


Figure 14. Implementation schedule

| Week   | Phase     | Deliverables                                                                                                                                             |
|--------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | Sprint 0  | Kick-off, backlog, UX flows, visual design, architecture approval, environments, legal pack outline                                                      |
| 2 to 3 | Sprint 1  | Mobile sign-in, owner onboarding, registries, manual items, image pipeline, dashboard, tenant configuration                                              |
| 4 to 5 | Sprint 2  | Guest pages, reservations, link completion, sharing and previews, escrow account creation, payout number verification                                    |
| 6 to 7 | Sprint 3  | Contributions, event processing, gift log, receipts and notifications, statements. <b>Milestone 2:</b> end-to-end demonstration with a live contribution |
| 8 to 9 | Sprint 4  | Automated payouts, commission, sweep, contribution limits, approvals, cancellation and refunds, premium                                                  |
| 10     | Sprint 5  | Administration console, exception queue, reports, consent and data requests, accessibility. <b>Milestone 3:</b> feature complete                         |
| 11     | Hardening | Security review, load test, UAT with pilot events, legal pack approval                                                                                   |
| 12     | Launch    | Production release, training, handover. <b>Milestone 4:</b> launch acceptance                                                                            |

Based on contract signature by 16 October 2026, work begins on 19 October 2026 and launch is scheduled for the week of 4 January 2027. Public holidays are accounted for.

## SECTION 14

# Testing and acceptance

| Test               | Objective                                                   | Exit criteria                                                           |
|--------------------|-------------------------------------------------------------|-------------------------------------------------------------------------|
| Automated tests    | Business rules and payment logic                            | All tests pass on every merge                                           |
| End-to-end tests   | Owner, guest, refund and premium journeys                   | Pass on Android Chrome, iOS Safari and desktop                          |
| Live payment tests | Contribution, payout, refund and sweep                      | KES 1 transactions reconcile with PayHero and the ledger                |
| Load test          | Peak traffic                                                | 2,000 contributions per hour and 50 page views per second within NFR-02 |
| Security review    | OWASP risks, authentication, payout changes, link retrieval | No open high or critical findings                                       |
| Accessibility      | WCAG 2.2 AA                                                 | Automated and manual checks passed                                      |
| User acceptance    | Business acceptance by Hadia and pilot users                | All M requirements accepted; no open severity 1 or 2 defects            |

## SECTION 15

# Commercial terms

## 15.1 Fixed price

| Work package                                                                                            | KES            |
|---------------------------------------------------------------------------------------------------------|----------------|
| Discovery, UX and visual design, architecture                                                           | 55,000         |
| Registry service: registries, items, reservations, link completion, gift log, statistics                | 135,000        |
| Web application: website, guest pages, owner dashboard, administration console                          | 140,000        |
| Funds integration: escrow, contributions, automated payouts, commission sweep, limits, refunds, premium | 115,000        |
| Mobile sign-in, message templates and preferences                                                       | 45,000         |
| Compliance features, legal pack drafts, accessibility                                                   | 55,000         |
| Testing, security review, UAT, launch and handover                                                      | 65,000         |
| <b>Total (inclusive of taxes)</b>                                                                       | <b>610,000</b> |

## 15.2 Payment schedule

| Milestone                   | Due            | Share | KES     |
|-----------------------------|----------------|-------|---------|
| M1 Contract signature       | Week 0         | 30%   | 183,000 |
| M2 End-to-end demonstration | End of week 7  | 30%   | 183,000 |
| M3 Feature complete         | End of week 10 | 30%   | 183,000 |
| M4 Launch acceptance        | End of week 12 | 10%   | 61,000  |

## 15.3 Monthly running costs

| Item                                                                   | KES per month           |
|------------------------------------------------------------------------|-------------------------|
| Hosting, storage, backups and Codevertex platform services             | 3,500 to 7,000          |
| Payment costs: PayHero service wallet, paybill or till, payout charges | 6,500 to 13,000         |
| SMS and WhatsApp messaging                                             | 3,000 to 6,000          |
| <b>Total</b>                                                           | <b>13,000 to 26,000</b> |

Messaging and payment costs vary with volume. Domain registration, legal fees and marketing are excluded.

## SECTION 16

# Risks and mitigations

| Risk                                             | Impact                          | Mitigation                                                                        | Owner             |
|--------------------------------------------------|---------------------------------|-----------------------------------------------------------------------------------|-------------------|
| PayHero onboarding exceeds five weeks            | Delayed live payments           | Initiated in week 1; reservation-only launch available as fallback                | Hadia, Codevertex |
| Regulatory requirement for payment authorisation | Delay or model change           | Legal opinion in weeks 2 to 6; capped holds and balances; instant payout option   | Hadia             |
| Fraudulent registries                            | Financial and reputational loss | Verified payout numbers, limits, first-payout hold, monitoring, moderation        | Hadia operations  |
| M-Pesa payout failures                           | Owner dissatisfaction           | Automatic reversal and retry, status enquiry, owner notification, exception queue | Codevertex        |
| Inconsistent retailer data                       | Slower item entry               | Manual entry as primary method                                                    | Codevertex        |
| Scope change                                     | Schedule slippage               | Change control; agreed exclusions                                                 | Both              |
| Holiday period availability                      | Slower decisions                | Holidays scheduled; weekly decision reviews                                       | Both              |
| Low cash gifting adoption                        | Revenue below expectation       | Pilots with planners and church networks; conversion analytics                    | Hadia             |

## SECTION 17

# Post-launch support

| Service      | Terms                                                                                            |
|--------------|--------------------------------------------------------------------------------------------------|
| Hypercare    | 30 days after launch; same-day response to payment incidents                                     |
| Warranty     | Defects against this specification corrected at no cost for 90 days                              |
| Monitoring   | Alerts on payout failures, reconciliation variances, error rates and latency                     |
| Handover     | Administration guide, finance guide, runbooks, legal pack, registers, architecture documentation |
| Enhancements | Native applications, Kiswahili and referral programmes quoted on request                         |

## SECTION 18

# Acceptance

Signature below confirms acceptance of the scope, schedule and price in this document. Changes are subject to the change control process in [Project delivery](#).

| Hadia     | Codevertex Africa Limited |
|-----------|---------------------------|
| Name      |                           |
| Title     |                           |
| Signature |                           |
| Date      |                           |